

Part A

1. The Company Concept Of One Person:

The innovatively solo enterprise is eminent by one person, called the Founder, who acts like both the owner, manager, and operator of the business (Wu et al., 2023). Diverse traditional businesses, which need more than one agency and workers, the successful venture uses digital tools in organizing the business (Rezazadeh & Bonehgazy, 2025). The firm uses artificial intelligence, automated computer software and agents to intensify the efficiency of humans in getting their normal tasks in the business project. New computerized platforms give the company the ability to manage many different attributes such as finances, marketing, and service functions by one complete collaboration process, thus make the Founder free from regular work and letting him focus on decision making while regular work is executed automatically (Rezazadeh & Bonehgazy, 2025).

2. One-Person Company Architecture:

In single operator architecture, the Founder is the final authority when decisions are made (Rezazadeh & Bonehgazy, 2025). In this system, the Founder will be dealing with many computerized AI agents where each one focuses on different area in the company (Sabarish, 2025). The computerized AI agents are software modules designed to do many different tasks, like the housekeeper in the home environment or the cash recorder in business that do money transactions. These agents are assigned to do different tasks by the Founder requests and questions. There are a lot of components in this system like the Founder and agents, the database that saves data, the task manager, and communication things that allow collaboration between the Founder and agents (MIT Sloan, 2025; Sabarish, 2025).

3. Multi-Agent Systems Roles:

A multi-agent system has a number of AI agents that work and collaborate together as a team to score a goal (K., 2026). In this matter, each individual AI agent will work like an independent entity, but they interact for effective collaboration of their activities to achieve the goal needed from each agent (K., 2026). For instance, the Finance Agent will be responsible for budgeting and cost, the Marketing Agent for the marketing activities, and the Customer Service Agent for listening to customers' problems. Even though each agent is a separate entity, collaboration among them will happen whenever necessary. Then, the whole process becomes organized and efficient. Using many AI agents for different operations in the business helps make hard operations to become possible, and this helps increase efficiency (MIT Sloan, 2025).

4. The Perspective Of Programming:

From the programming perspective, the one-man company model can be made as a form of object-oriented programming (Leite, 2006). In OOP, the development of the system is done using classes and objects representing real things in the real world (Sabarish, 2025). For example, the founder might be programmed as a class that has attributes such as name and the ability to make decisions, whereas the agents could be treated as other classes assigned to do tasks. The tasks are objects themselves and have more info about the task. In the use of objects, interfaces, inheritance, encapsulation, and polymorphism, the written code can be stylish and reusable. An agent or the set of agents will be inherited from the main "Agent" parent class, but each agent will show its own special behavior (Leite, 2006; Rezazadeh & Bonehgazy, 2025).

Part B

1. Name of the Business:

We named our business “Smart Study Tutoring Services”.

2. Background of the Business:

Smart Study Tutoring Service is an online tutoring company where we focus on quality teaching to give good studying skills for the students. This business gives tutoring services for science, math, and programming subjects. The services are now available to those students who want to study over the internet (Online). The records are kept digitally in databases. Our vision is to provide and give quality tutoring services at an inexpensive cost to all students, and it could be managed by an individual easily.

3. Target Customers:

The target list for this product are students from various schools and universities who want help in their studies. It would help people who are preparing for exams, those who find difficulties in specific areas, and the students who wish to enhance their academic performance. Parents are also from our target market who want additional education for their kids. Since this is an online platform, it can reach all customers across the globe.

4. How One-Person Can Deal With This Business:

Our project operates in digital ecosystems and uses automated agents, so one person can control everything. Activities like scheduling classes, sending reminders and collecting payments and fees can be done automatically using AI agents. The Founder spends most of their time teaching, making decisions, and monitoring the system. With the help of AI agents, activities that need more time can be done without human participation. So it's possible to do the entire business operation at home using only one person.

5. Our Agents:

The system will have several agents coded to do its job:

- Finance Agent: Deals with the collection of fees and payments.
- Marketing Agent: It helps marketing teachers' requests and needs on social media apps and other websites.
- Operations Agent: It does the scheduling and task allotment of tutors.
- Customer Support Agent: It handles students' questions, reminds them, and gives important help to them.

Each agent acts as an expert in each domain of the company.

6. How Founder And AI Agents Interact In This Business:

The Founder sends the requests and questions to the AI agents. For example, the Founder may ask the Operations Agent to book tutoring for a student. The Marketing Agent can come up with new innovative ideas on how to attract and bring more students that need help in light of existing trends. The Finance Agent provides the Founder with the every monthly earnings report, and the Customer Support Agent informs the Founder about the questions asked by students. After sending everything to the Founder it falls onto his shoulders to make the final decision regarding service, pricing and session changes.

Part C

1.The Architecture Of Our System:

Our business system is built to use a Multi-Agent system so it can be a one-person company, where the main decision-maker will be the Founder. The system uses a modular design pattern, so the Founder designs the task, then the different AI agents do the task. These agents have the domain knowledge and can generate Recommendations; however, the final decision will be on the Founder's shoulders.

2. Structure of Multi-Agents:

- The Finance Agent: It will collect fees and will track all receipts.
- The Marketing Agent: It will deal with advertisements and manage social media apps to inductee more students.
- The Operations Agent: It will control scheduling and conducting.
- The Customer Support Agent: It will answer student questions.

3. System Classes:

The above system architecture uses this essential classes for the code implementation:

-Founder class:

It's for the actor who can create tasks and receive reports from agents.

-Agent class/es:

It's the behavior of the digital AI agents and the requirements.

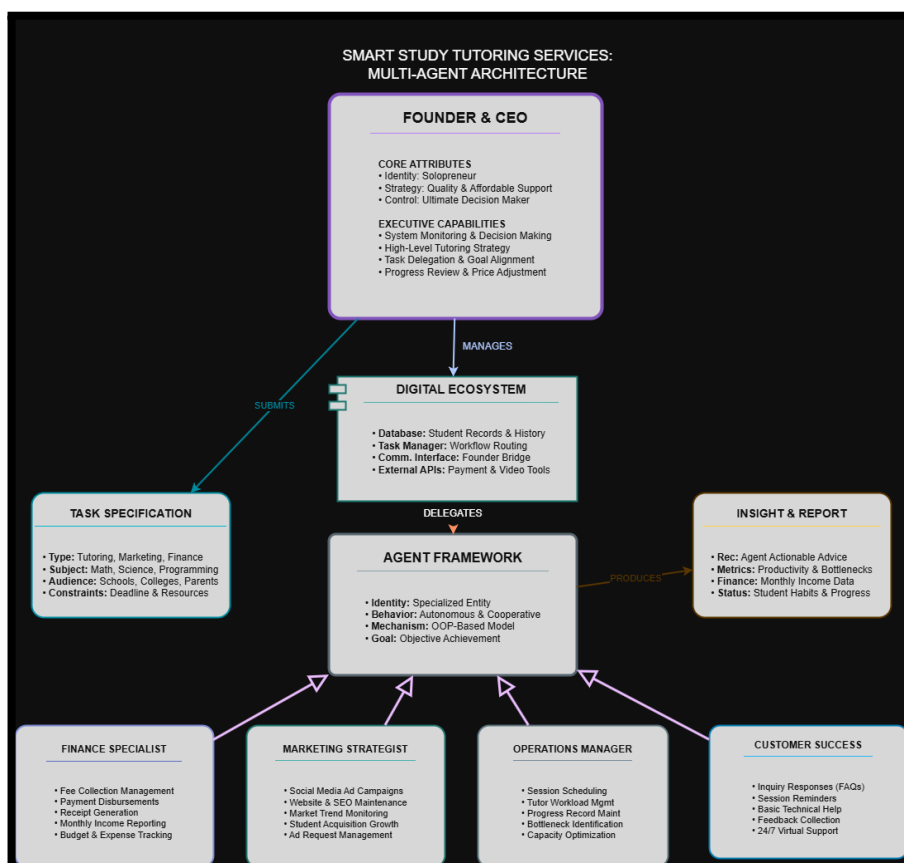
-Task object:

It represents actions made by the founder in the business.

-Reports-AND-Recommendation class:

It's the output created by the AI agent after processing the data received as a task.

4.The Diagram Of Class As an UML:



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